Existence as Stability and Expansion: A Systemic Epistemic Definition Across Biological, Socio-Political, and Consciousness-Based Frames

Kevin T. N.
Independent Researcher
jkdkr2439@gmail.com

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I wrote this to search for a coherent logic that fits my own experiential observations and internal explanatory structure. References to existing literature are included **only as connective points**—to relate, contrast, and communicate ideas—not as the source from which the framework was derived. In other words, this is structurally different from "reading theories and then summarizing them"; the explanatory scaffold preceded the citations, and the citations serve as links, not as foundations.

I do not claim this document is **the truth**, or even **the best model**. It is a provisional **positioning step**—a ladder for orientation—and it may change substantially with further discussion or evidence. If parts overlap with ideas proposed elsewhere, that should be understood as convergence in structural reasoning rather than any claim of novelty or "reinventing the wheel".

If you would like to discuss, critique, or extend these ideas, please contact: jkdkr2439@gmail.com.

Abstract

This paper-formatted note proposes a system-level epistemic definition of *existence* as the joint requirement of *stability* and *expansion*. In biological systems, this dual requirement appears as *survival* and *development*, where development is defined as expanding the boundary conditions of survival so that survival can continue under broader constraints. Survival is operationalized as the capacity to absorb and assimilate flows from the environment (energy, matter, and information) in a way that maintains structure. In socio-economic and political systems, the same invariant is reframed as stability of operation and expansion of controllable boundary conditions (resources, legitimacy, and coordination range). In consciousness-oriented or energy-grounded systems, stability is treated not as static persistence but as *resonant coherence*: a dynamically maintained attractor-like pattern that persists through coupling. The paper develops the definition, clarifies frame-dependence, and marks common sources of category errors (e.g., conflating physical energy with functional activation). The goal is to provide a compact conceptual scaffold for an epistemology series rather than a completed scientific theory.

Keywords: epistemology; systems; existence; stability; expansion; survival; development; resonance; coherence; boundary conditions.

1 Introduction

In many domains, "existence" is treated as a self-evident primitive: things either exist or do not. This note adopts a different stance: *existence is defined operationally as a system-level invariant* that can be recognized across multiple frames. The motivation is epistemic rather than metaphysical: to build a reusable scaffold for reasoning about systems—biological, socio-political, and consciousness-oriented—without requiring an absolute ontology.

The central claim is simple:

> *To exist (as a system) is to maintain stability while enabling expansion.*

The remainder of the paper makes this claim precise, then translates it into three application frames (biology, socio-political systems, and consciousness/energy-grounded systems). Throughout, the paper treats definitions as *frame-bound* and *relational*: the same invariant can appear under different vocabularies depending on what the system counts as a boundary, a resource, or a failure mode.

2 Conceptual Framework

2.1 System, environment, and boundary conditions

A *system* is taken to be a pattern that persists by maintaining constraints on its own dynamics despite environmental perturbations. The environment provides both enabling flows and destabilizing noise. Existence, in this sense, is a *relational* property: it is always defined *with respect to* a system–environment boundary.

2.2 Two invariants: stability and expansion

We define two coupled requirements.

Stability (self-maintenance). Stability is the capacity to maintain an identifiable organizational pattern under perturbation. Stability can be static (a fixed structure) or dynamic (a maintained attractor), depending on the frame.

Expansion (boundary extension). Expansion is the capacity to extend the range of conditions under which stability can be maintained. Expansion is not optional "growth"; it is a strategy for continued stability in changing or competitive environments.

Existence (systemic definition).

$$\text{Existence} \triangleq \text{Stability} + \text{Expansion}. \tag{1}$$

The plus sign is not arithmetic; it indicates joint necessity in the intended sense: stability without expansion is brittle, while expansion without stability collapses. The coupling resembles classic cybernetic intuitions about regulation and viability (Ashby 1956; Bertalanffy 1968).

2.3 A note on frame-dependence and category errors

A recurring failure mode in cross-domain reasoning is *frame conflation*: carrying a term from one frame into another as if it retained the same referent. The most common case in this note is the word "energy". In physical frames, energy is a conserved quantity; in functional/cognitive frames, the same word often denotes activation, arousal, motivation, or attentional intensity. This note allows functional uses but flags them explicitly to avoid metaphysical drift.

3 Biological Frame: Survival and Development

3.1 Existence as survival + development

In the biological frame, the systemic definition is expressed as:

$$\textit{Existence} \approx \textit{survival} + \textit{development},$$

with a specific constraint:

Development is the expansion of survival boundaries so that survival can continue.

This makes development instrumentally subordinate to survival, not as a moral claim but as an operational one: development is what prevents the system from being trapped in a narrow niche.

3.2 Survival as absorption and assimilation

Survival is operationalized as the capacity to *absorb* and *assimilate* flows from the environment:

- **Energy flow:** metabolic intake and conversion.
- **Material flow:** nutrients, building blocks, and waste regulation.
- **Information flow:** sensing and learning that improves regulation.

The key point is not the substance absorbed, but the *maintenance of organizational constraints* through managed throughput. This view is broadly consonant with autopoietic intuitions: a living system maintains itself as a self-producing network of processes (Maturana and Varela 1980).

3.3 Development as boundary expansion

Development expands the set of viable conditions by:

- improving regulatory control (better homeostasis/allostasis),
- widening accessible resource regimes (new niches, new behaviors),
- increasing predictive and adaptive capacity (learning and plasticity).

In this frame, existence is not merely persistence; it is persistence *plus* the capacity to remain persistent under broader and more uncertain conditions.

4 General Systems Frame: Stability and Expansion in Socio-Economic and Political Systems

4.1 Translation of invariants

In socio-economic and political systems, "survival" is less biological but still meaningful if reframed as *system viability*:

- **Stability:** continuity of institutional operation, coordination, and conflict management.
- **Expansion:** extension of controllable boundary conditions (resources, influence, legitimacy, and coordination range).

This translation preserves the invariant while changing the measurement vocabulary.

4.2 What counts as absorption here

The biological notion of absorption becomes *resource acquisition and assimilation*:

- **Economic:** capital flows, productive capacity, supply-chain continuity, and trust/credit.
- **Political:** legitimacy, compliance, enforcement capacity, and institutional coherence.

Stability is maintained by keeping the operational constraints intact; expansion reduces future fragility by widening accessible resources and influence.

4.3 Stability–expansion coupling and failure modes

The coupling can fail in two symmetrical ways:

- **Stability without expansion:** brittle stagnation; vulnerability to shocks and competitors.
- **Expansion without stability:** overextension; loss of coherence; collapse of coordination.

This mirrors the original definition: existence requires both poles to remain jointly satisfied.

5 Consciousness/Energy-Grounded Frame: Stability as Resonant Coherence

5.1 Why stability is not static here

In systems whose "substrate" is primarily dynamical (oscillatory, activity-based, or process-based), "stability" often means *a maintained pattern* rather than a fixed structure. Stability is therefore described as *resonance* or *coherent coupling*.

5.2 Resonant stability as an attractor-like pattern

In this frame:

Stability ≈ resonant coherence (a dynamically maintained pattern).

Resonant coherence refers to the system's ability to keep its internal relations aligned despite perturbation. In neural terms, coherence is frequently discussed as phase alignment that enables effective inter-areal communication (Fries 2015). This note uses "resonance" in the same functional spirit: it is a shorthand for *stable relational alignment in time*.

5.3 Expansion as widening the resonance bandwidth

Expansion is then reframed as:

Expansion ≈ widening the domain of coherent maintenance.

Informally: the system can sustain coherent patterns across more contexts, higher complexity, or greater noise. In consciousness-oriented language, this can correspond to broader integrative capacity and more robust control under uncertainty. Predictive-processing styles of thinking treat robust experience and action as arising from continual prediction–error regulation (Friston 2010), which is compatible with the present framing: expansion increases the range of conditions under which regulation remains coherent.

5.4 Common category error: physical energy vs functional activation

A specific warning applies here. If "energy" is taken literally as a physical conserved quantity, claims about "consciousness as energy" become metaphysical quickly. If "energy" is taken functionally (activation, arousal, intensity, coupling strength), the claims become operational but frame-bound. This note adopts the functional usage and treats "resonance" as a descriptor of maintained coupling, not as a statement about exotic physics.

6 Discussion

6.1 What this definition provides

The proposed definition provides:

- a compact invariant (stability + expansion) that can be translated across domains,
- a principled way to interpret "development" as boundary expansion of viability,
- a clean reframing of dynamical stability as resonance/coherence in process-based systems,
- a built-in guardrail against conflating vocabularies across frames.

6.2 Why this is epistemology, not ontology

The note is intentionally epistemic: it offers a way to *organize* observations and reasoning, not a final claim about what ultimately exists. This stance is consistent with using references as connectors rather than foundations. The definition can be judged by its coherence, reuse across frames, and its ability to reduce confusion in reasoning.

6.3 Limitations and next steps

Limitations include:

- lack of quantitative measures for stability and expansion in each frame,
- unresolved mapping between functional "resonance" and specific measurable variables,
- potential ambiguity in where to draw the system boundary in socio-political contexts.

Next steps (for the planned series) include formalizing stability/expansion as measurable proxies within each frame, and providing worked examples that demonstrate error-checking when frames collide.

7 Conclusion

This paper-formatted note defines existence (as a system-level invariant) as the joint requirement of stability and expansion. In biological systems, this appears as survival and development, with development defined as expanding survival boundaries to maintain survival. In socio-economic and political systems, the invariant appears as operational stability and expansion of controllable boundary conditions. In consciousness- or energy-grounded systems, stability is reframed as resonant coherence: a dynamically maintained pattern that persists through coupling, while expansion widens the domain in which coherence can be sustained. The definition is offered as an epistemic scaffold for continued refinement and dialogue rather than as a scientific claim of final truth.

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